Statistical Consulting Portfolio

Selected Projects

CARE Consortium: Head Impact Exposure and Concussion in NCAA Football
Professor / Assistant Dean of Research Analytics, School of Public Health (2020–2022)

- Conducted rigorous data wrangling and validation of 700,000+ head-impact records, correcting duplicate records, data-entry problems, and systematic UTC time-conversion errors.
- Developed 4 R Shiny apps for descriptive statistics, model selection, and network visualization.
- Conducted survival analysis to assess the relationship between prolonged impact exposure and concussion diagnosis.
- Applied hierarchical clustering to match concussed players with comparable counterparts and conducted functional data analysis to evaluate group differences.

Development of Data Management Systems
Executive Director, Biostatistics Consulting Center (2021–2022)

COVID-19 Sample Tracking System
- Developed an R Shiny app to monitor a cloud database and track sample collection progress.

Research Project Management System
- Revised and debugged an R Shiny app for accessing cloud data storage.
- Developed a custom pie-donut plotting function to replace a package incompatible with R Shiny.

RStudio Connect (Posit Connect) Implementation
- Collaborated with the IT team to test system functionality, including package compatibility, cloud data management, and authentication.

Health-Organization Network Construction and Interactive Visualization
Associate Professor, Department of Applied Health Science (2017 & 2020)

Hospital and Health-Organization Partnerships
- Designed an algorithm to construct a graph adjacency matrix from survey responses.
- Developed an R Shiny app to visualize organizational partnership networks.

COVID-19 Organizational Partnerships
- Designed partnership network structures from questionnaire data.
- Developed an R Shiny app to interactively generate networks based on reported partnership strength.

Clinical Trial Statistical Code Review and Multiple-Imputation Validation
Postdoctoral Fellow, Department of Epidemiology and Biostatistics (2018)

- Reviewed R code for a generalized linear mixed model comparing behaviors across three groups.
- Cross-validated multiple-imputation analyses for missing data between SAS and R.
- Developed custom R code for post-hoc GLMM analysis with multiply imputed data beyond the functionality available in the *mice* package.

Smoking Cessation Trials in Pregnant Women
Associate Professor, Department of Applied Health Science (2017)

- Participated in model-selection discussions for longitudinal data from 300+ participants.
- Recommended a random-slope mixed-effects model based on individualized within-subject changes over time.
- Developed an animated R Shiny app to visualize individual trajectories and flag missing data.

Additional Projects

Statistical Analysis & Modeling

Breast Cancer Risk Communication Survey — data cleaning, model selection, logistic regression. *(Pharmacy, 2019)*
Demographic Differences in Standardized Test Scores — exploratory analysis, permutation-based pseudo-longitudinal pairing. *(Science Education, 2015)*
Sports Financial Metrics and International Medal Counts — ANOVA and Shiny visualization of sports financial metrics and Olympic medal performance. *(Sport Management, 2015)*
Security Configuration User Study — non-parametric test recommendations, publication figures. *(Computer Science, 2014)*
Historical Change in Word Order — model selection for Bayesian logistic mixed-model analysis using JAGS. *(Computational Linguistics, 2014)*
Revenue Cycle Management Evaluation — time-series transformation and custom plotting function for 100+ variables. *(Higher Education, 2013)*

R Shiny Applications

Human and Dog Activity Tracking — cloud-based R Shiny data tracking system for two activity-monitoring devices. *(Applied Health Science, 2019)*

Indiana Cancer Risk Mapping — interactive R Shiny + Leaflet app visualizing colon cancer risk geographically. *(Biostatistics Consulting Center, 2016)*
Custom 3D Surface Visualization — convolution algorithm and R Shiny app for mesh resolution, color, and peak controls. *(Biology, 2016)*
Transparency and Openness Promotion Guidelines Survey — reusable R Shiny template for research data applications. *(Epidemiology, 2020)*

Network Data Visualization

Vendor–Supplier Networks in an Emerging African City — network and geographic visualization. *(Anthropology, 2020)*
Video-Viewing Behavioral Network Analysis — adjacency matrix from Jaccard similarity, network plots of behavioral clusters. *(Media Psychology, 2014)*

Data Processing, Code Review & Validation

Accelerometer Data Processing — SAS/R data merging, activity quantification, systematic-error handling. *(Epidemiology & Biostatistics, 2020)*
Circadian Protein Expression Study — reproduced 100+ GraphPad Prism figures with a custom R plotting function, reviewed ANOVA Type III code. *(Public Health, 2020)*
Randomized Crossover Trial of a Decaffeinated Energy Drink — GLMM code review, SAS/R Bayes-factor cross-validation. *(Public Health, 2019)*
R Package Debugging (CCTpack) — identified a data-transposition bug causing incorrect clustering results. *(Anthropology, 2018)*

Methods at a Glance

- Generalized linear mixed models & longitudinal models (*lme4, nlme*)
- Survival analysis (*survival*)
- Functional data analysis (*refund*, *fda*)
- Network analysis (*network*, *statnet*, *igraph*)
- Missing data & multiple imputation (*mice*)
- Data visualization & mapping (*ggplot2*, *leaflet*)
- R Shiny app development (*shiny, bslib*)